

End-Term BTP Presentation on

“MILP Heuristics in 3D Bin Packing Problem”

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Literature review and Research gap

- The **3D Bin Packing Problem (3D-BPP)** is a strongly NP-hard problem with wide applications in logistics, supply chain management, and packaging. It involves arranging rectangular SKUs into fixed-dimension bins while minimizing bin usage or maximizing volume utilization.
- **Pre-Mid term Approach:-** Heuristic Methods: Implemented three heuristic algorithms Layer-Based (horizontal layer stacking), Peak Filling Slice Push (gap identification and filling), and Space Defragmentation (item repositioning). While computationally faster, these heuristics scale poorly for large heterogeneous datasets with many SKUs, often producing suboptimal packings due to greedy placement strategies.
- **Exact Methods (MILP):** The formulations provide optimal solutions through branch-and-bound algorithms but face severe scalability issues due to combinatorial explosion. For instances with 7+ SKUs, CPLEX solvers typically timeout before finding optimal solutions.



Literature review and Research gap

- **Column Generation & Decomposition Methods:** Studied Dantzig-Wolfe decomposition from "Introduction to Linear Optimization" (Bertsimas & Tsitsiklis, 1997) and foundational cutting stock work by Gilmore & Gomory (1961).
Column generation iteratively generates variables (patterns) dynamically rather than enumerating all possibilities upfront, enabling solution of large-scale linear programs with exponentially many variables.
- **Cutting Stock Problem & 2D Packing:** 1D cutting stock (Gilmore-Gomory) minimizes stock waste; 2D cutting stock extends this to rectangular piece cutting from rectangular sheets. Literature (Lodi et al., 2002) documents column generation with Master Problem (pattern selection) and Pricing Problem (pattern generation via knapsack subproblem).

Research Gap Addressed: Limited practical implementation of Delayed Column Generation for 3D bin packing problems. Earlier heuristic methods showed poor performance (1.5-2.1% efficiency). This work implements 2D DCG as a systematic foundation—demonstrated success on 2D cutting stock can be extended to 3D geometric constraints, bridging the gap between decomposition theory and packing applications.



Problem Statement/ BTP Objectives

- Minimize unused volume in 3D bin packing with constraints on item orientation, weight balance, and bin dimensions.
- MILP solvers scale poorly while heuristics give inconsistent results.
- **Pre-Midterm Objective (3D Bin Packing):** Minimize total unused bin volume across all bins used for packing items.
- **Post-Midterm Objective (2D Cutting Stock via DCG):**
minimize the total number of stock sheets required to satisfy all item demands

$$\begin{aligned} &\text{minimize} && \sum_{p \in P'} x_p \\ &\text{subject to} && \sum_{p \in P'} a_{ip} x_p \geq d_i \quad \forall i \in I \\ &&& x_p \geq 0 \quad \forall p \in P' \end{aligned}$$

Objectives

- Develop automated logging for solver performance tracking
- Study and implement Delayed Column Generation extending from 1D to 2D cutting stock
- Formulate and validate 2D DCG with correct Master and Pricing problem constraints
- Establish foundation for future 3D bin packing extension



Research Methodology

Phase 1: Pre-Midterm

Heuristic Development:

- Implemented Layer-Based, Peak Filling Slice Push, and Space Defragmentation heuristics in Python (numpy, pandas, itertools)
- Integrated with IBM ILOG CPLEX using docplex interface; heuristic solutions used as warm-start initialization
- Tested on real picklist data (3-10 SKUs) from warehouse systems
- Findings: Heuristics showed poor standalone performance (1.5-2.1% efficiency); CPLEX baseline often outperformed heuristic-initialized runs

Phase 2: Post-Midterm

- Implemented automated logging system capturing timestamps, picklist IDs, solution status, objective values, optimality gaps, branch-and-bound node counts, and final variable values
- Conducted intensive study of Delayed Column Generation theory; mastered Dantzig-Wolfe decomposition principles (took 2 weeks due to conceptual complexity; recovered through accelerated implementation)
- Successfully formulated and implemented complete 2D DCG solver with validated correctness and visualization.



2D Delayed Column Generation - Mathematical Formulation:

Restricted **Master Problem** (Pattern Selection):

$$\text{Minimize } \sum_{p \in P'} x_p$$

Subject to:

$$\sum_{p \in P'} a_{i,p} \cdot x_p \geq d_i \quad \forall i \in I \quad (\text{demand satisfaction})$$

$$x_p \geq 0 \quad \forall p \in P'$$

x_p = number of sheets cut using pattern p

$a_{i,p}$ = count of item type i in pattern p

d_i = demand for item type i

P' = current restricted pattern set

Pricing Problem (Subproblem):

$$\text{Maximize } \sum_{i \in I} p_i \cdot a_i$$

Where p_i = dual prices from Master Problem; a_i = number of items of type i in new pattern

Subject to geometric packing constraints:

- Containment: All items must fit within sheet bounds $L \times W$
- No overlap: Enforced via binary relative position variables: left-right, above-below
- Conflict prevention: An item cannot be simultaneously left AND right of another (same for above/below)
- Item count: Exactly a_i items of each type i must be placed in the pattern

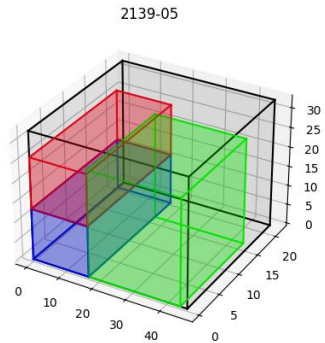
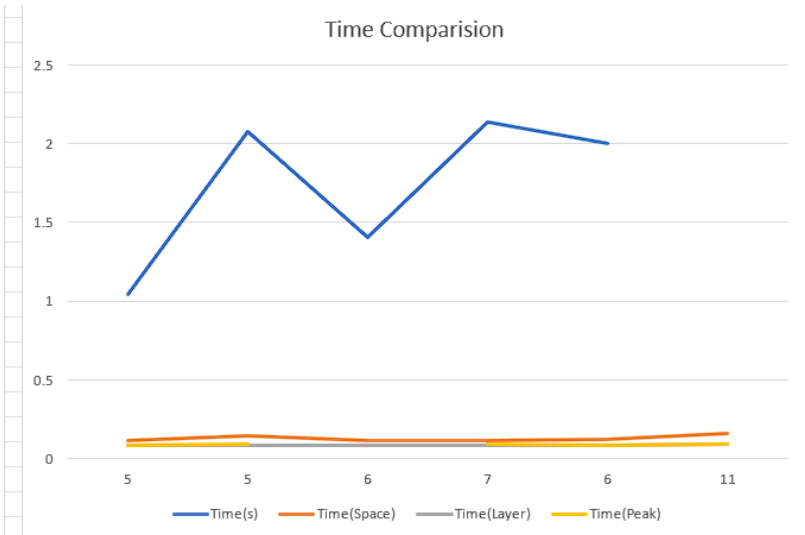
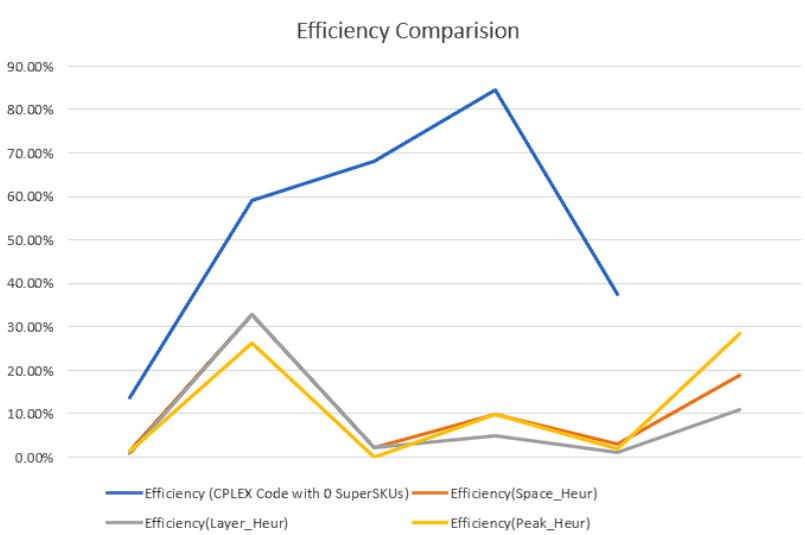
Termination: If reduced cost $c_p = 1 - \sum_i p_i \cdot a_{i,p} \geq -\epsilon$, STOP (optimal); else add pattern to P' and repeat.



Results and Discussion(Pre-Midterm Heuristic Performance)

- Hybrid MILP + Heuristics
 - Main CPLEX MILP model alone often matched or outperformed heuristic-initialized runs.
 - Visualization: matplotlib 3D plots of bins with wireframes and coloured cuboids.

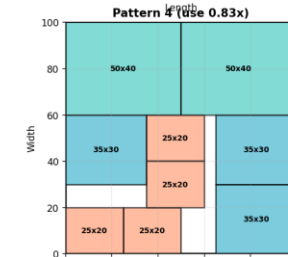
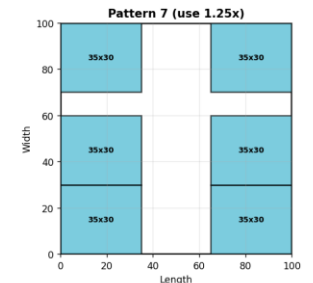
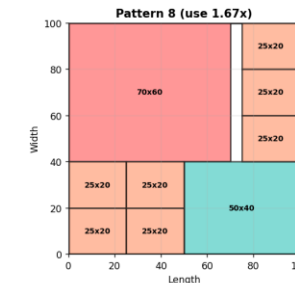
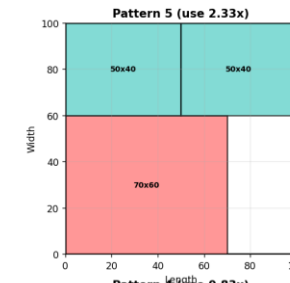
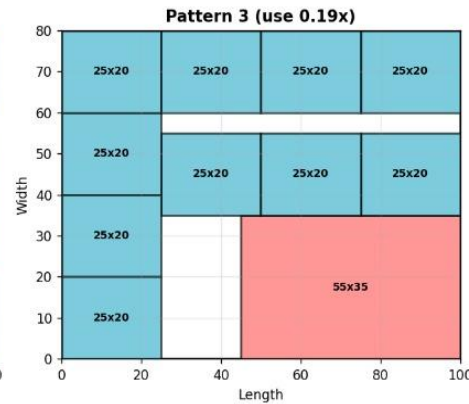
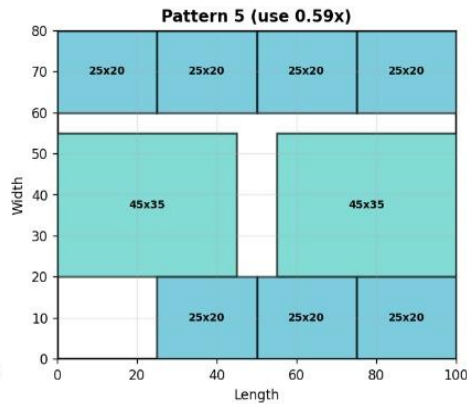
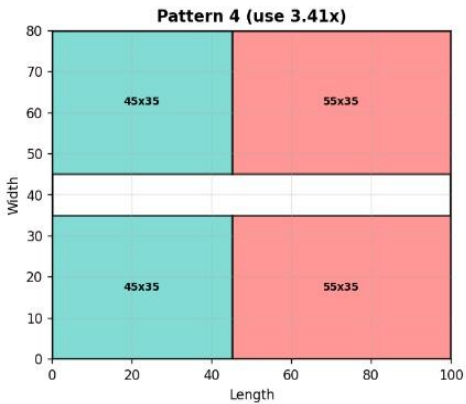
Picklist Barcode	Qty	Efficiency (CPLEX Code with 0 SuperSKU)				Efficiency (CPLEX Code with Layer_He)			Efficiency (CPLEX Code with Peak_heu)			Efficiency (CPLEX Code with Space_He)			Efficiency (Space_H Time(Spa)			Efficiency(Lay			Efficiency (Peak_He Time(Pea			Number of SKUs
		Time(s)	s	Value	ur	ur	r	ur	ur	ur	ur	Value(Space)	eur	ce	ur	Value(Layer)	er_Heur	Time(Layer)	ur	ur	ur	ur	ur	ur
809098706	1	1.04	13.54%	22410	Time Limit	13.54%	13.54%					239538	1.44%	0.119		482586	0.72%	0.088		239538	1.44%	0.087		5
ALLIED-1	5	2.075	59.04%	55157	Time Limit	59.04%	59.04%					1635.48	32.71%	0.146		1635.48	32.71%	0.088		179448	26.17%	0.092		5
809580564	6	1.402	68.03%	2526	68.03%	68.03%	68.03%					237672	2.21%	0.117		237672	2.21%	0.087		NA				6
809103772	1	2.136	84.38%	4401	84.38%	84.38%	84.38%					219274	9.78%	0.119		462322	4.89%	0.088		219274	9.78%	0.089		7
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Results and Discussion(Post-Midterm: 2D DCG Implementation)

Implementation Success & Validation:

- Successfully implemented complete 2D DCG solver with CPLEX pricing problem
- Verified on a few test cases: demand satisfaction, pattern feasibility, no-overlap constraints, optimality (reduced costs $\geq -\epsilon$)
- Developed 2D matplotlib visualization with color-coded item types, dimensions, and sheet utilization



• Sheet: 100 x 80
Demands: {(55, 35): 7, (45, 35): 8, (25, 20): 6}
Total sheets (LP): 4.13
Total sheets (rounded): 5

Sheet: 100 x 100
Demands: {(70, 60): 4, (50, 40): 8, (35, 30): 10, (25, 20): 15}

Total sheets (LP): 6.08
Total sheets (rounded): 7



Phase 1: Logging System for 3D MILP Model:

- Implemented FileProgress listener capturing node-level MIP progress: best incumbent, best bound, optimality gap, solve time, node counts.
- This was to wrap up the previous work done and to store logs for the instances of SKUs which didn't generate a solution.

Phase 2: 2D DCG Pattern Generation Results:

- **Example Patterns:** Generated varied cutting patterns showing item placement optimization. Pattern 9 (use 3.21x) packs items efficiently with single 70×60 item and multiple 50×40 items; Pattern 4 (use 3.41x) combines 45×35 and 55×35 items in balanced layout. Sheet utilization ranges 79-88% across test instances.
- **Key Finding:** DCG produces compact, diverse patterns with high sheet utilization; successful validation demonstrates correct constraint formulation and pricing problem optimization for 2D cutting stock scalability.



Conclusions

Two-Phase Journey:

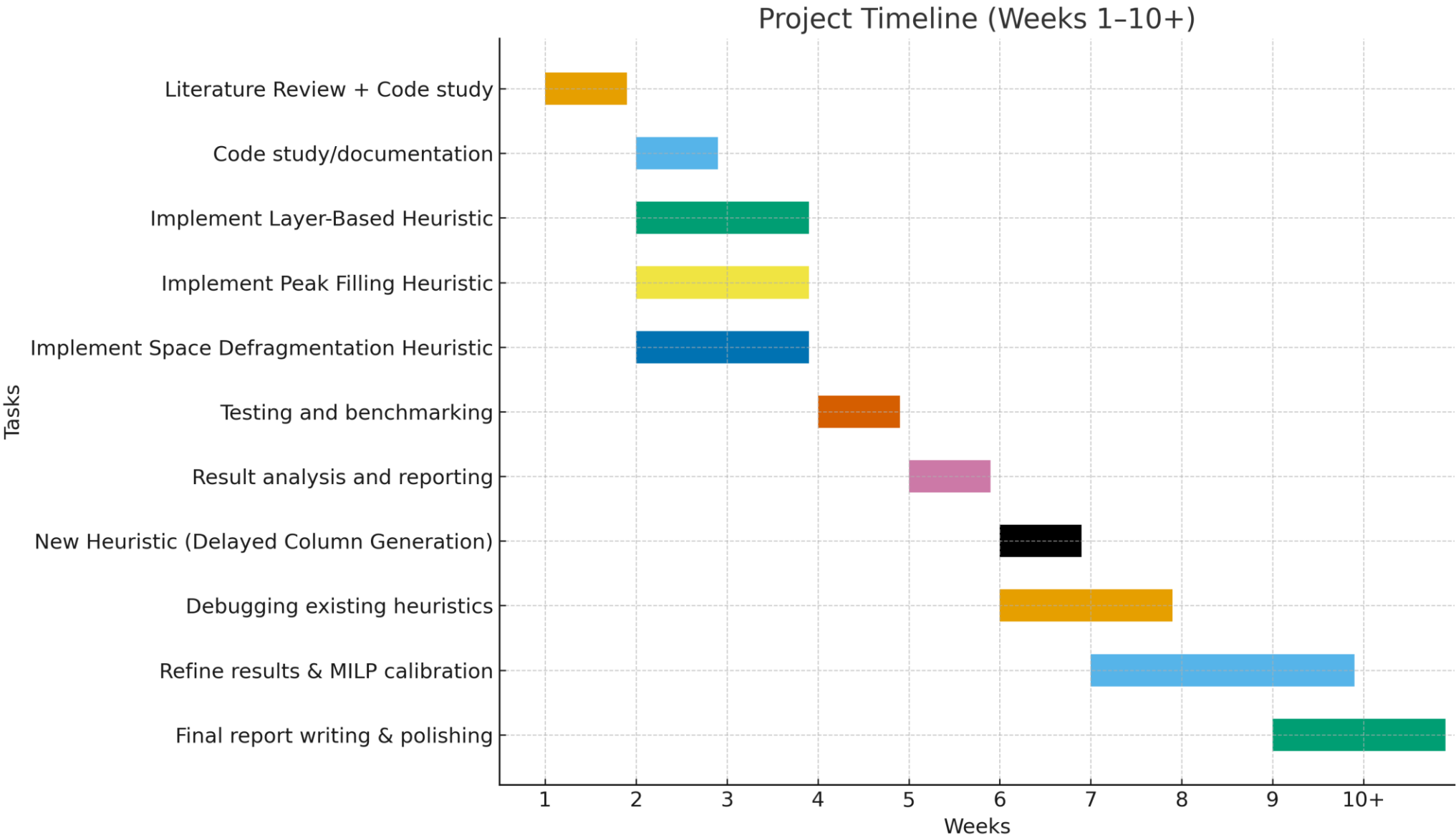
Progressed from implementing heuristic algorithms for 3D bin packing through MILP integration to developing systematic decomposition techniques via Delayed Column Generation.

Key Learning Insights:

- Infrastructure & Remote Computing: Learnt tmux for sessions on high-performance servers (Atreyee)
- Logging & Debugging: Implemented robust logging frameworks for tracking solver behavior across timeouts and failures; automated performance profiling enabled rapid identification of bottlenecks.
- Current Achievement: Working 2D DCG implementation with validated correctness—stable foundation for future 3D extension by adding height dimension to geometric constraints.
- Future Work:
Extend 2D DCG to 3D geometric constraints; implement 3D visualization; develop heuristic pricing accelerators for computational efficiency; integrate branch-and-price for integer solutions.



GANTT Chart



Key Deviation:
DCG
theoretical
study consumed
2 weeks (Week
7-8) instead of
planned 1 week
due to the
learning curve.
Recovered by
accelerating
implementation
post-Week 10
breakthrough
with validation
on test cases.

